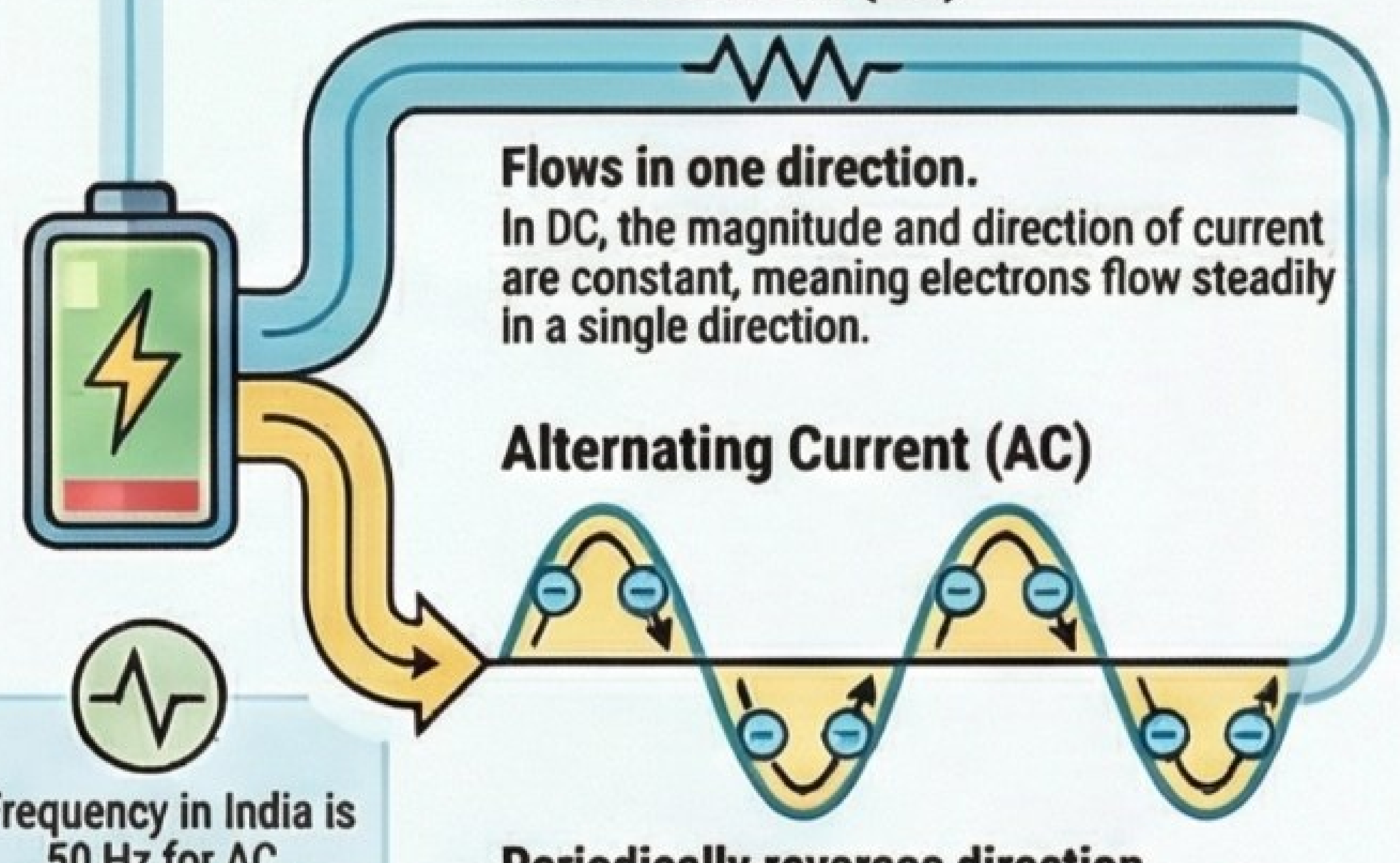
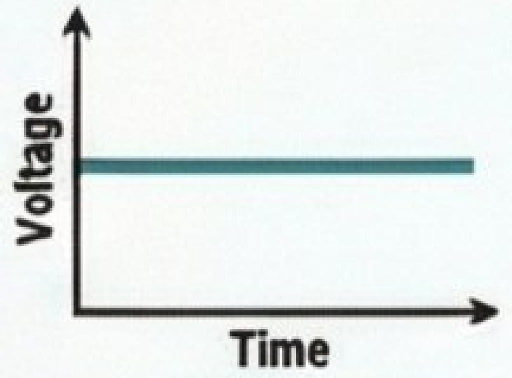
ALTERNATING CURRENT: A VISUAL REVISION GUIDE

AC vs. DC: THE FUNDAMENTALS



Direct Current (DC)

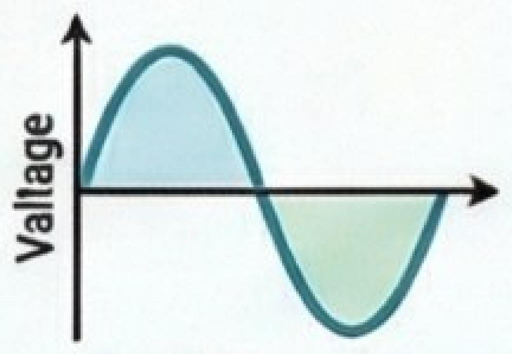
Flows in one direction.
In DC, the magnitude and direction of current are constant, meaning electrons flow steadily in a single direction.



Waveform Shape: Straight Line

Alternating Current (AC)

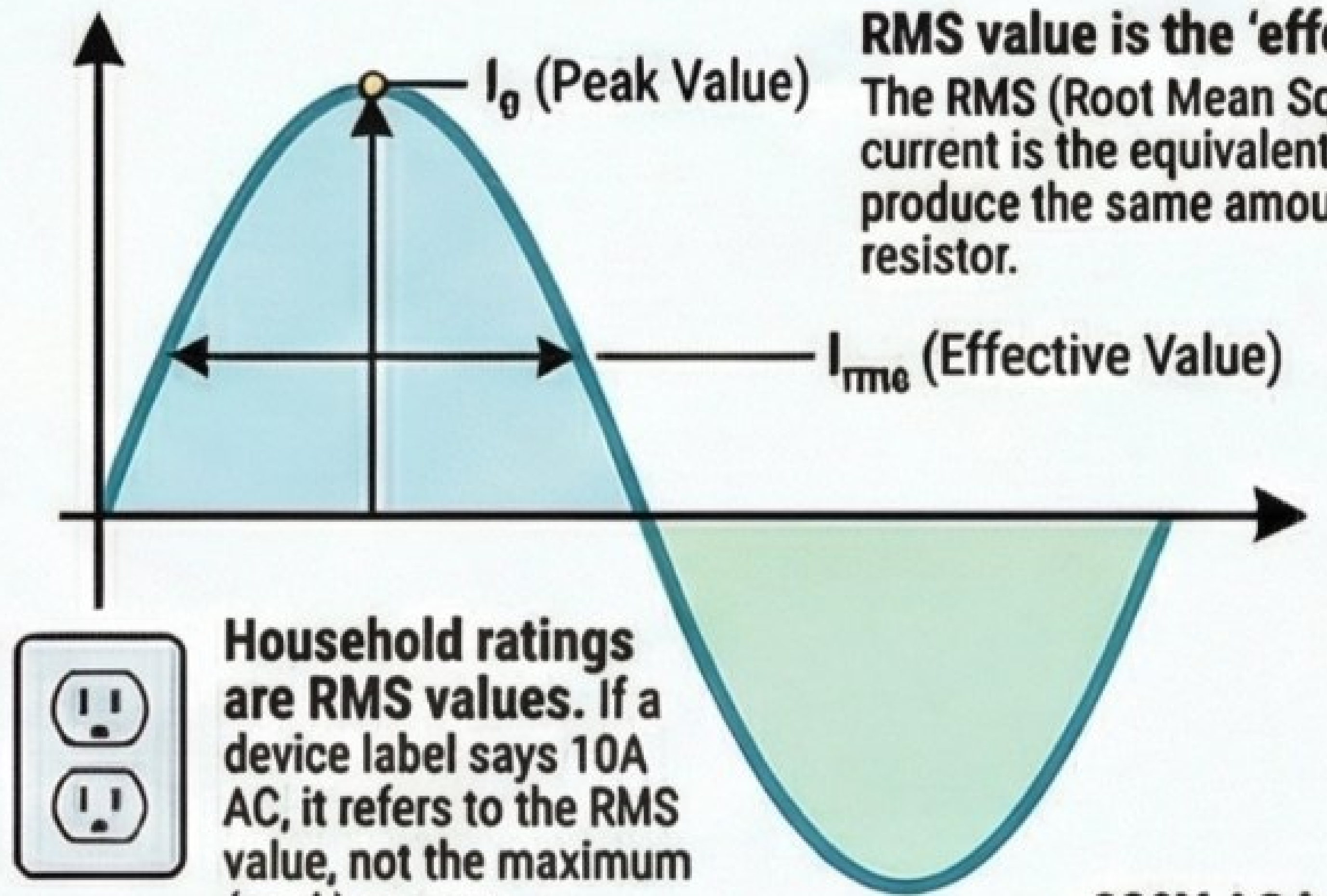
Periodically reverses direction.
in AC, electrons oveillate back and forth. The magotiods changes continuously, while the direction reverses after a fixed time interval.



Waveform Shape: Sine Wave

Frequency in India is 50 Hz for AC, This means one, the frequency of DC is always 0 Hz.

UNDERSTANDING AC VALUES: PEAK vs. RMS



RMS value is the 'effective' value of AC.
The RMS (Root Mean Square) value of an AC current is the equivalent DC current that would produce the same amount of heat in the same resistor.

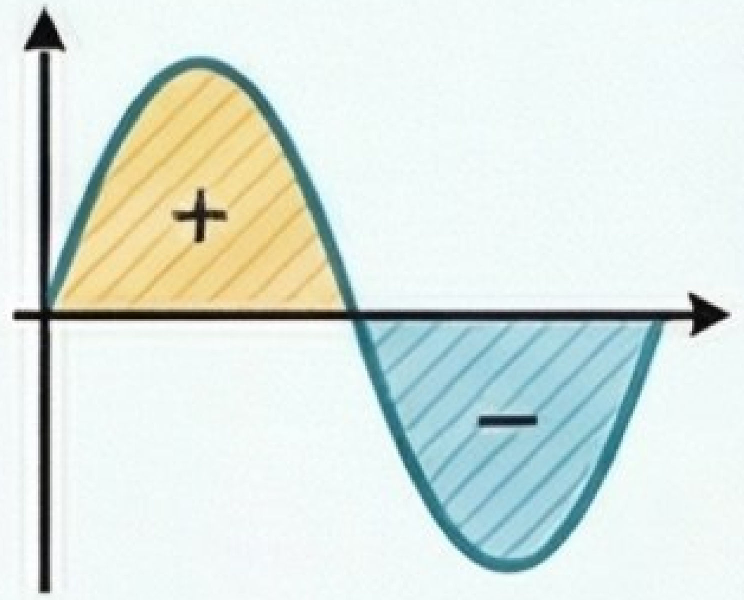
Household ratings are RMS values. If a device label says 10A AC, it refers to the RMS value, not the maximum (peak) current.

Relating RMS and Peak Values:

$$I_{rms} = \frac{I_0}{\sqrt{2}}.$$

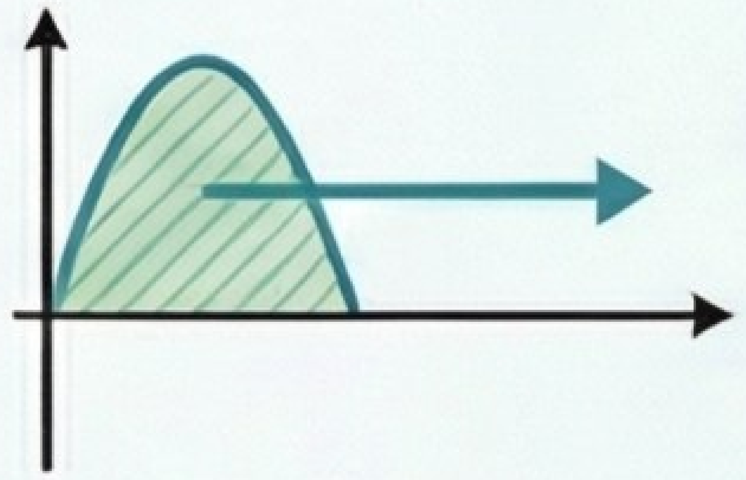
220V AC is more dangerous than 220V DC. A 220V DC shock is constant at 220V. For 220V AC (which is the RMS value), the peak voltage reaches 220V = ৩2 = 311V, delivering a more severe shock.

KEY AVERAGES & DERIVATIONS



Average current over a full cycle is ZERO.
The positive half-cycle cancels out the negative half-cycle, resulting in a net average value of zero.

This is why AC doesn't produce a chemical effect.



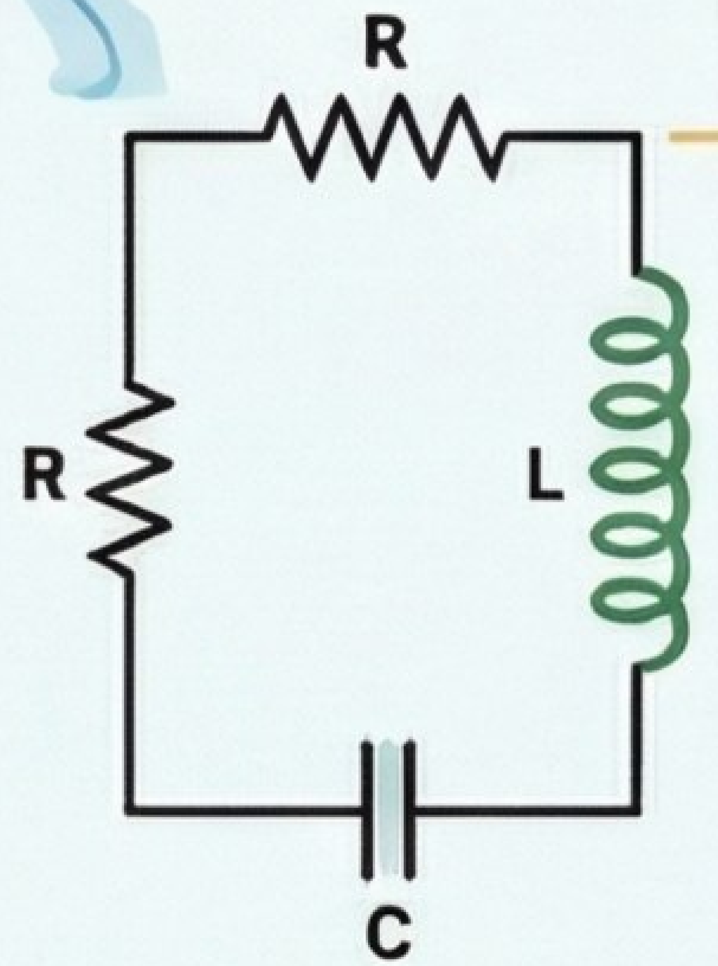
Average current over a half cycle is not zero.

Formula: $I_{avg} = 0$

The total charge transferred over one complete cycle is zero.

ESSENTIAL AC DEVICES & CIRCUITS

LCR CIRCUIT & IMPEDANCE



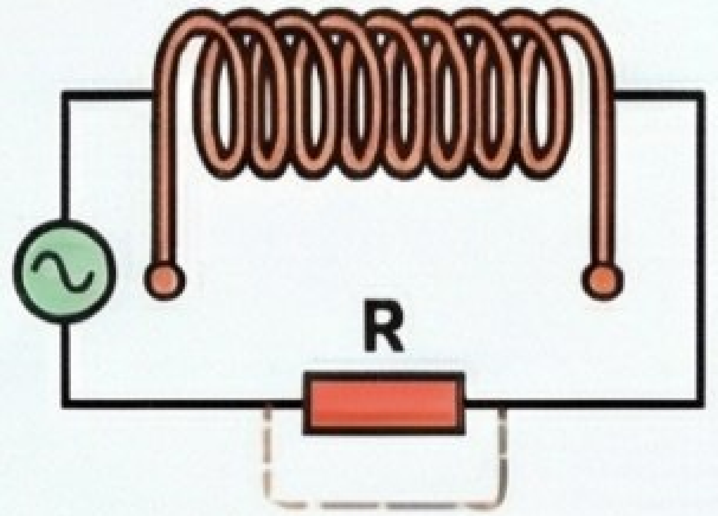
In a series circuit with a resistor (B), inductor (L), and capacitor (C), the total opposition to current is called **impedance** (Z).

$$Z = \sqrt{R^2 + (X_L - X_D)^2}$$

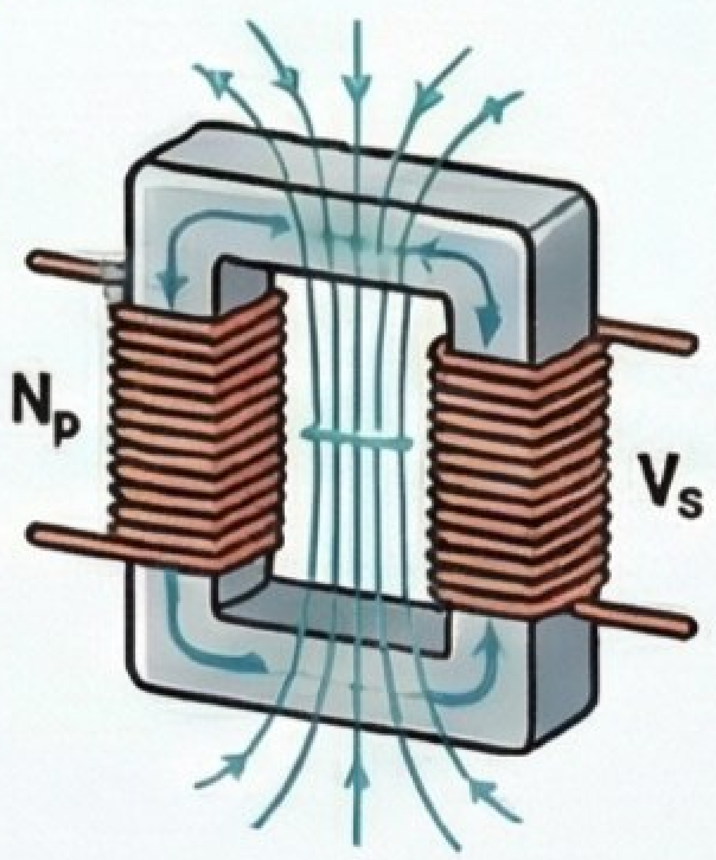
Electrical Resonance:
Resonance occurs when the circuit's impedance is at its maximum (Z=N), allowing maximum current to Row. This happens when inductive reastance (X_L) equals capacitive reastance (X_C).

CHOKE COIL

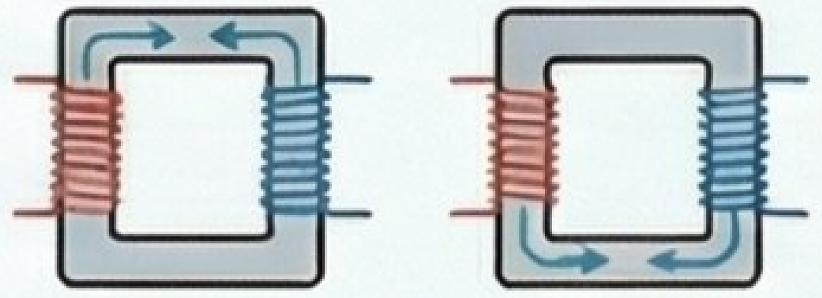
Controls current efficiently.
A choke cpll is an inductor used to control AC with minimal power loss, as it opposes current via self-induction rather than dissipating energy as heat like a resistor.



TRANSFORMER



Changes AC voltage levels.
Working on the principle of mutual induction, a transformer can increase (step-up) or decrease (step-down) AC voltage. It does not work with DC.



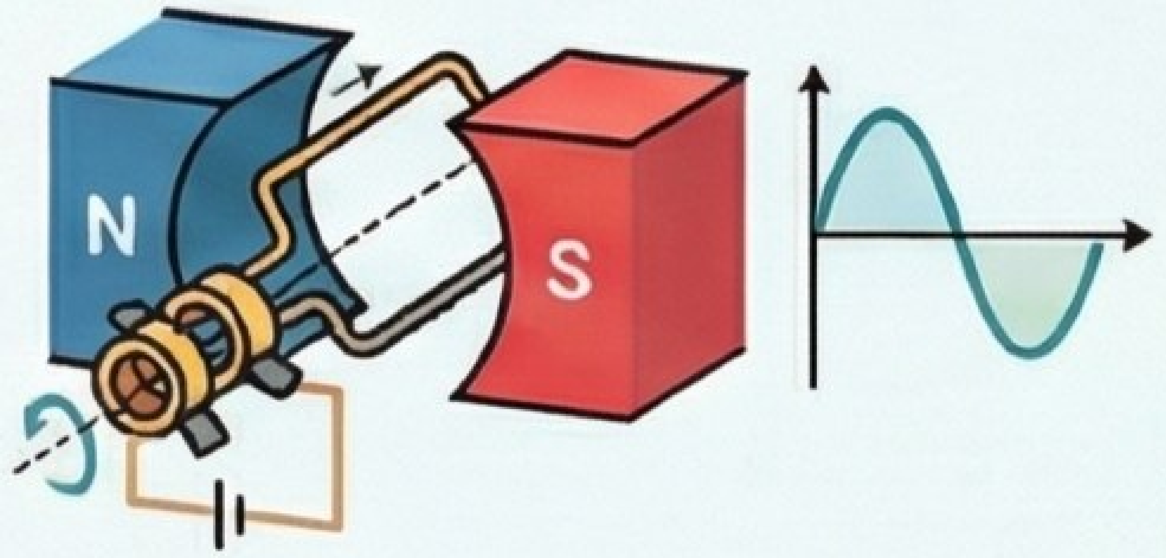
Step-Up: $N_b > N_p$, Voltage Increase

Step-Down: $N_b < N_p$, Voltage Decrease

Transformers conserve power.
In an ideal transformer, input power equals output power ($V_pI_p = V_oI_o$).

AC GENERATOR

Converts mechanical to electrical energy.
Based on the principle of electromagnetic induction, It generates AC by rotating a coil in a magnetic field, which continuously changes the magnetic flux.



How it Works: As the coil rotates, the direction of the induced current (determined by Fleming's Right-Hand Rols) reverses every half rotation, creating alternating current.